

```
operations on lists
```

```
list1=list(1,2,3)
```

```
list2=list("water","juice","lemonade")
```

```
list1
```

```
1. 1
2. 2
3. 3
```

```
list2
```

```
1. 'water'
2. 'juice'
3. 'lemonade'
```

Merging lists

```
list12=c(list1,list2)
```

```
list12
```

```
1. 1
2. 2
3. 3
4. 'water'
5. 'juice'
6. 'lemonade'
```

List to vector

```
list1 = list(1,2,3)
```

```
list2 = list("water","juice","lemonade")
```

```
unlist(list1)
```

```
1 2 3
```

```
unlist(list2)
```

```
'water' 'juice' 'lemonade'
```

```
mode(list1)
```

```
'list'
```

```
mode(unlist(list1))
```

```
'numeric'
```

Appending lists

```
list1 = list(1,2,3)
list2 = list("water","juice","lemonade")
```

```
append(list1,100)
```

```
1. 1
2. 2
3. 3
4. 100
```

```
append(list2, "coffee")
```

```
1. 'water'
2. 'juice'
3. 'lemonade'
4. 'coffee'
```

```
list1 = list(1,2,3)
list2 = list("water", "juice", "lemonade")
```

```
append(list1, 100, after=2)
```

```
1. 1
2. 2
3. 100
4. 3
```

```
append(list2, "coffee", after=2)
```

```
1. 'water'
2. 'juice'
3. 'coffee'
4. 'lemonade'
```

Removing from list

```
list1 = list(1,2,3)
list2 = list("water", "juice", "lemonade")
```

```
list1[-2]
```

```
1. 1
2. 3
```

```
list2[-1]
```

```
1. 'juice'
2. 'lemonade'
```

Extracting from list

```
list1 = list(1,2,3,4,5,6)
list2 = list("water", "juice", "lemonade", "tea", "coffee", "milk")
```

```
list1
```

```
1. 1
2. 2
3. 3
4. 4
5. 5
6. 6
```

```
list1[2:4]
```

```
1.2
2.3
3.4
```

```
list1[c(1,3,5)]
```

```
1.1
2.3
3.5
```

```
list2
```

```
1. 'water'
2. 'juice'
3. 'lemonade'
4. 'tea'
5. 'coffee'
6. 'milk'
```

```
list2 = list("water", "juice", "lemonade", "tea",
"coffee", "milk")
```

```
list2
```

```
1. 'water'
2. 'juice'
3. 'lemonade'
4. 'tea'
5. 'coffee'
6. 'milk'
```

```
list2[2:4]
```

```
1. 'juice'
2. 'lemonade'
3. 'tea'
```

```
list2[c(1,3,5)]
```

```
1. 'water'
2. 'lemonade'
3. 'coffee'
```

Vector Indexing

```
x = 1:10
```

```
x
```

```
1 · 2 · 3 · 4 · 5 · 6 · 7 · 8 · 9 · 10
```

```
x[ (x > 5) ]
```

```
6 · 7 · 8 · 9 · 10
```

```
x[ (x%%2==0) ]
```

```
2 · 4 · 6 · 8 · 10
```

```
x[ (x%%2==1) ]
```

```
1 · 3 · 5 · 7 · 9
```

Vector Indexing

```
x[5] = NA
```

```
x
```

```
1 · 2 · 3 · 4 · <NA> · 6 · 7 · 8 · 9 · 10
```

```
y = x[ !is.na(x) ]
```

```
y
```

```
1 · 2 · 3 · 4 · 6 · 7 · 8 · 9 · 10
```

```
mean(x)
```

```
<NA>
```

```
mean(y)
```

```
5.555555555555556
```

□ Vector of negative integers

```
x = 1:10
```

```
x
```

```
1 · 2 · 3 · 4 · 5 · 6 · 7 · 8 · 9 · 10
```

```
x[-(1:5)]
```

```
6 · 7 · 8 · 9 · 10
```

```
x[(6:10)]
```

```
6 · 7 · 8 · 9 · 10
```

□ String vector

```
z = list(a1 = 1, a2 = "c", a3 = 1:3)
```

```
z
```

```
$a1
1
$a2
'c'
$a3
1 · 2 · 3
```

```
names(z)
```

```
'a1' · 'a2' · 'a3'
```

```
z = list(a1 = 1, a2 = "c", a3 = 1:3)
```

```
names(z)[3] = "c2"
```

z

```
$a1
  1
$a2
'c'
$c2
  1 · 2 · 3
```

```
x = c(water=1, juice=2, lemonade=3 )
```

```
names(x)
```

```
'water' · 'juice' · 'lemonade'
```

```
x["juice"]
```

```
juice: 2
```

□ Empty index

```
x = 1:10
```

x

```
1 · 2 · 3 · 4 · 5 · 6 · 7 · 8 · 9 · 10
```

```
x[]
```

```
1 · 2 · 3 · 4 · 5 · 6 · 7 · 8 · 9 · 10
```

□ Mixed mode

```
ab = list(1, 2, 3, "X", "Y", "Z")
```

```
dim(ab) = c(2,3)
```

```
print(ab)
```

```
      [,1] [,2] [,3]
[1,]  1    3  "Y"
[2,]  2    "X"  "Z"
```

```
mode(print(ab))
```

```
      [,1] [,2] [,3]
[1,]  1    3  "Y"
[2,]  2    "X"  "Z"
'list'
```

Factors

```
factor(x)
```

```
1 · 2 · 3 · 4 · 5 · 6 · 7 · 8 · 9 · 10
▼ Levels:
'1' · '2' · '3' · '4' · '5' · '6' · '7' · '8' · '9' · '10'
```

```
help("factor")
```

◆ Gemini

```
factor(x, levels = unique(x))
```

```
1 · 2 · 3 · 4 · 5 · 6 · 7 · 8 · 9 · 10
```

```
► Levels:
```

```
y = c(1, 4, 3, 5, 4, 2, 4)
```

```
possible.dieface = c(1, 2, 3, 4, 5, 6)
```

```
labels.dieface = c("one", "two", "three", "four", "five", "six")
```

```
facy = factor(y, levels = possible.dieface, labels = labels.dieface)
```

```
facy
```

```
one · four · three · five · four · two · four
```

```
▼ Levels:
```

```
'one' · 'two' · 'three' · 'four' · 'five' · 'six'
```

```
y = c(1, 4, 3, 5, 4, 2, 4)
```

```
y
```

```
1 · 4 · 3 · 5 · 4 · 2 · 4
```

Factors – Class and Unclass

```
x = c(3, 4, 5, 6, 1, 2, 3, 3, 4, 4, 5, 6)
```

```
x
```

```
3 · 4 · 5 · 6 · 1 · 2 · 3 · 3 · 4 · 4 · 5 · 6
```

```
y=as.factor(x)
```

```
y
```

```
3 · 4 · 5 · 6 · 1 · 2 · 3 · 3 · 4 · 4 · 5 · 6
```

```
▼ Levels:
```

```
'1' · '2' · '3' · '4' · '5' · '6'
```

```
x=factor(c("lemonade", "lemonade", "juice", "lemonade", "water"))
```

```
x
```

```
lemonade · lemonade · juice · lemonade · water
```

```
► Levels:
```

```
class(9)
```

```
'numeric'
```

```
class("9")
```

```
'character'
```

```
class(print)
```

```
'function'
```

```
x = matrix(nrow=2, ncol=2, data=1:4)
```

```
class(x)
```

```
'matrix' · 'array'
```

```
brands = c("A", "A", "B", "B", "B", "B", "C")
```

```
brands
```

```
'A' · 'A' · 'B' · 'B' · 'B' · 'B' · 'C'
```

```
brands_fac = factor(brands)
```

```
brands_fac
```

```
A · A · B · B · B · B · C
```

► **Levels:**

```
unclass(brands_fac)
```

```
1 · 1 · 2 · 2 · 2 · 2 · 3
```

```
colours = c("blue", "green", "red")
```

```
colours
```

```
'blue' · 'green' · 'red'
```

```
unclass(brands_fac)
```

```
1 · 1 · 2 · 2 · 2 · 2 · 3
```

```
colours [unclass(brands_fac)]
```

```
'blue' · 'blue' · 'green' · 'green' · 'green' · 'green' · 'red'
```

```
x = factor( c( "lemonade", "lemonade", "juice", "lemonade", "water" ) )
```

```
unclass(x)
```

```
2 · 2 · 1 · 2 · 3
```

```
x = factor( c("lemonade", "lemonade", "juice", "lemonade", "water"),  
levels=c("water", "juice", "lemonade") )
```

```
x
```

```
lemonade · lemonade · juice · lemonade · water
```

► **Levels:**

```
unclass(x)
```

```
3 · 3 · 2 · 3 · 1
```

```
levels(x)
```

```
'water' · 'juice' · 'lemonade'
```

Factors Example for an ordered factor:

```
income = ordered(c("high", "high", "low", "medium", "medium"), levels=c("low", "medium", "high") )
```

```
income
```

```
high · high · low · medium · medium
```

► **Levels:**

```
unclass(income)
```

```
3 · 3 · 1 · 2 · 2
```

Strings – Display and Formatting

Print and Format Function

```
print( sqrt(2) )
```

```
[1] 1.414214
```

```
print( sqrt(2), digits=5 )
```

```
[1] 1.4142
```

```
print( sqrt(2), digits=10 )
```

```
[1] 1.414213562
```

```
print("apple")
```

```
[1] "apple"
```

```
print(c("apple", "banana"))
```

```
[1] "apple" "banana"
```

```
print(c("apple", "banana", 6, 10))
```

```
[1] "apple" "banana" "6" "10"
```

```
print( format( 0.5, digits=10, nsmall=15 ) )
```

```
[1] "0.500000000000000"
```

```
format(c("A", "BB", "CCC", "DDD"),
width = 7, justify = "centre")
```

```
' A ' BB ' CCC ' DDD '
```

```
format(c("A", "BB", "CCC", "DDD"),
width = 14, justify = "centre")
```

```
' A ' BB ' CCC ' DDD '
```

Formatting and Display of Strings: Use of justify

```
format(c("A", "BB", "CCC", "DDD"), width = 7, justify = "centre")
```

```
' A ' BB ' CCC ' DDD '
```

```
format(c("A", "BB", "CCC", "DDD"), width = 7, justify = "left")
```

```
'A ' BB ' CCC ' DDD '
```

```
format(c("A", "BB", "CCC", "DDD"), width = 7, justify = "right")
```

```
' A ' BB ' CCC ' DDD '
```

```
format(c("A", "BB", "CCC", "DDD"), width = 7, justify = "none")
```

```
'A ' BB ' CCC ' DDD '
```



```
x = matrix(nrow=3, ncol=2, data=1:6, byrow=T)
```

```
print(x)
```

```
      [,1] [,2]  
[1,]    1    2  
[2,]    3    4  
[3,]    5    6
```

Formatting and Display of Large Quantities

```
format(1234567, big.mark = ",")
```

```
'1,234,567'
```

```
format(12345678, big.mark = ",")
```

```
'12,345,678'
```

```
format(123456789, big.mark = ",")
```

```
'123,456,789'
```

```
format(123456789, big.mark = " ")
```

```
'123 456 789'
```

END